Athanasios (Thanos) Klidaras

Remote sensing geologist • Reflectance spectroscopy and mineral mapping
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Geologist with experience in NASA Perseverance rover science operations, geological mapping, and experience in identifying aqueous alteration minerals with VNIR-SWIR reflectance spectroscopy. Combines Python and GIS analysis with leadership of scientific working groups and spacecraft operations experience. Available to work upon PhD completion (expected May 2027).

Relevant experience

Purdue University | NASA Perseverance rover | Mastcam-Z science operations team *2022–present*

- Conceptualized scientific investigation campaigns, gathered science-team support, and secured approval to conduct campaigns with the Perseverance rover; delivered >70 presentations to the mission science team.
- Leadership positions: Co-lead the Geologic Context Working Group and the Multispectral Working Group. Dozens of preeminent geoscientists from the Perseverance science team gather for these weekly meetings.
- Analyzed multispectral observations of igneous and sedimentary in Jezero crater to investigate mineral composition and geological context, with a focus on fluvial deposits and aqueous alteration.
- Trained for Mastcam-Z science operations in downlink (time-sensitive analysis of recent data) and uplink (planning of data acquisition) roles.

Purdue University | Mars Data Analysis Program *2022–2025*

- Analyzed visible and near-infrared (VNIR) orbital spectroscopy, high-resolution imagery, and digital terrain models to investigate clay stratigraphy and constrain ancient surface environments and climates on Mars.
- Identified aqueous alteration minerals at 90 sites, focused on kaolinite, smectite, iron oxides, acid sulfates.
- Performed spectral and geospatial data analysis and visualization in Python; published first-author findings in Geology. Collaborators: B. Horgan, W. Farrand, R. Navarre, T. Goudge, S. Olson, R. Wordsworth

Geology fieldtrips

2023 University of Florida-led GeoSPACE trip to northern Arizona: San Francisco volcanic field, Grand Canyon, meteor crater
2022 Imperial College-led trip to central Italy: structural geology of the Apennines fold-and-thrust belt and Elba ophiolite
2019 Imperial College-led trip to NW Scotland: 2 weeks of sedimentary logging in ~1.0 Ga Torridon Group rocks

Technical skills

- **Geology:** Field and image-based geological mapping, sedimentary logging, igneous and alteration mineral identification, and petrological descriptions.
- **Remote sensing and GIS:** Processing and interpretation of terrestrial (Landsat, ASTER, WorldView) and planetary (CRISM, Mastcam-Z, Magellan SAR) remote sensing datasets in ArcGIS, ENVI, and Google Earth Engine.
- **Spectroscopy:** Multispectral and hyperspectral reflectance data at visible (VIS), near-infrared (NIR), and shortwave-infrared (SWIR) wavelengths. Trained to use ASD FieldSpec & QualitySpec, HySpex, Specim-IQ field/lab spectrometers.
- **Programming and analysis:** Python, NumPy, IDL, and MATLAB for spectral and geospatial analysis; data visualization and statistical analysis (bootstrapping, uncertainty quantification).
- **Operations:** Mastcam-Z science downlink and uplink training; familiarity with United States government procurement, security, and export controls (regular ITAR training).
- **Technical communication:** Scientific presentations, mapping reports, and graphics for technical and public audiences; GIMP and Inkscape.

Education

Purdue University, West Lafayette, IN, USA *Aug 2022 – present*
GRADUATE STUDENT IN PLANETARY SCIENCE
Advisor: Prof Briony Horgan
Passed qualifying exams in 2023, passed preliminary exams in 2025, planning PhD defense in spring 2027

Imperial College, London, UK *Sep 2018 – Jul 2022*
INTEGRATED MASTERS (MSci) IN GEOLOGY *[First-Class Honours]*
Advisor: Dr Philippa Mason
Thesis: Mapping recent activity in Atla Regio, Venus: The tectono-magmatic evolution of Ozza and Maat Montes
BACHELOR OF SCIENCE (BSc) IN GEOLOGY *[First-Class Honours]*
Independent mapping project: The Tectonostratigraphic Evolution of the Eastern Kalpin Fold-and-Thrust Belt

Awards

2025	***Purdue University (EAPS)***	**H. Jay Melosh Planetary Conference Support**	**$1407**
2025	***Lunar & Planetary Institute***	**LPI Career Development Award**	**up to $1000**

2025	*Purdue University (EAPS)*	Expo – Best Student Poster	$100
2024	*NASA*	Group Achievement Award – Mars2020 Jezero western fan & margin units team	N/A
2023	*Purdue University (EAPS)*	Research Blitz - Best Student Presentation	N/A
2022	*Imperial College (RSM)*	The Watts Medal - Top academic performance in Geology	N/A
2021	***Imperial College (RSM)***	**E. Ambrosiadou Prize - Top academic performance in geologic mapping project**	**$1300**
	Imperial College (RSM)	Dean's List	N/A
2020	*Imperial College (RSM)*	Dean's List	N/A

Technical teaching

Undergraduate teaching assistant | Imperial College *2021*

- Optical Mineralogy: trained students to use a petrological microscope, identify minerals, and write whole-rock petrological descriptions.
- Independent Mapping: trained students to digitize field maps in ArcGIS, write mapping reports, and produce conference-quality posters.

Teaching assistant | Purdue University *2025*

- Remote Sensing: trained students in IDL coding, GIS analysis, and spectral analysis

Science communication and outreach

- Mars Perseverance Blog contributor (2023–present). Contributed "A Striped Surprise" (September 2024), which received over 100 million views and generated dozens of international news articles, including coverage in Forbes.
- Invited speaker, Harrow School (September 2020). Delivered a 40-minute talk on lunar geology, following work with Prof Jan Cilliers, to an audience of 80 in person and online.
- Moderate r/Space, a community of 27 million subscribers; hosted a virtual community meet-up with former astronaut Chris Hadfield. Create science essays with millions of views and tens of thousands of interactions, and educational posters displayed in schools and observatories.

Publications

Google Scholar | ORCID

Select journal publications

2026

- H. T. Manelski, C. Bedford, Dong Jae Lee… **A. Klidaras** et al, (2026). LIBS Elemental Mapping of a Feldspathic Lunar Meteorite: Insights for In Situ Exploration, *JGR: Planets*, doi:10.1029/2026JE009769
- A. J. Jones, R. Barnes, S. Gupta… **A. Klidaras** et al, (2026). A Fluvio-Lacustrine Environment Preserved in the Jezero Crater Inlet Channel, Neretva Vallis, Mars, *JGR: Planets*, doi:10.1029/2025JE009420

2025

- **A. Klidaras**, R. Navarre, B. H. Horgan, W. Farrand, T. Goudge, R. Moore. Global Topography of Clay Deposits Signify An Epoch of Warm and Humid Climate on Early Mars, *Geology,* doi:10.1130/G52988.1
- P. J. Mason, **A. Klidaras**, D. Cirium, R. Ghail, S. Lea-Wurzbach. Evolution of Plume Volcanism at Atla Regio, Venus, *JGR: Planets,* doi:10.1029/2024JE008815
- K. H. Williford, K. A. Farley, B. Horgan… **A. Klidaras** et al, (2025). Carbonated ultramafic igneous rocks in Jezero crater, Mars, *Science*, doi:10.1126/science.adu8264

2024

- C. Royer, C. C. Bedford, J. R. Johnson… **A. Klidaras** et al, (2024). Intense alteration on early Mars revealed by high-aluminum rocks at Jezero Crater, *Communications Earth & Environment*, doi:10.1038/s43247-024-01837-2

Select first author conference abstracts

Oral presentations

- **A. Klidaras**, B. Horgan, A. Broz, T. Goudge, W. Farrand, C. Albright, S. Olson (2026). Kaolinite Signatures Reveal Variable Weathering Intensity Across Early Mars, *57th LPSC*, abstract #1540
- **A. Klidaras**, B. H. Horgan, A. F. Vaughan, M. Rice, J. R. Johnson, J. I. Simon, B. Orenstein, B. S. Kathir, D. T. Flannery, J. F. Bell III (2024). Diversity of Fluvially Transported Clasts in the Jezero Western Fan Using Mastcam-Z, *Tenth International Conference on Mars*, abstract #3223
- **A. Klidaras** and P. J. Mason (2022). Mapping Recent Activity on Venus: The Tectono-Magmatic Evolution of Ozza Mons and the History of Atla Regio, *53rd LPSC*, abstract #2678

Poster presentations

- **A. Klidaras**, R. Navarre, B. H. Horgan, W. Farrand, T. Goudge, R. Moore (2024). Global Topography of Clay Deposits Signify An Epoch of Warm and Humid Climate on Early Mars, *55th LPSC*, abstract #1516